BIO-RAD

Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, suspension

Species Used Mouse, erythroleukemia cells

Molecules Electroporated DNA: pSV2 neo, 5 kB

Before the Pulse

Cell Growth Medium RPMI ,10% Fetal Calf Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest Log phase

Pre-pulse Incubation 10 to 20 min.

Wash Solution Not given

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 4 °C

Electroporation Medium* Not given

Cell Density 10 (7) cells / 50 µl

Volume of Cells 50 µl

DNA Concentration Not given

DNA Resuspension Buffer Not given

Volume of DNA Not given

Cuvette Gap 0.4 cm

Voltage 0.4 kV

Field Strength 1 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) none Ω

Time Constant 10 msec

After the Pulse

Outgrowth Medium RPMI,10% Fetal Calf Serum

Outgrowth Temperature 37 °C

Length of Incubation 10 to 30 minutes

Selection Method or Assay Used

Electroporation Efficiency Low (**see notes)

Per Cent Survival 20 to 70%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

**Note: Transfection efficiemnces ofthese cells may be increased with the use of the Capacitance Extender (providing longer time constants).

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